NETWORK AND PHYSICAL SECURITY

Questions and Answers

This assignment explains common network threats, demonstrates a safe offline PIN-guessing simulation, identifies physical risks to server rooms, and presents practical safeguards for a college computer lab.

Question 1. Common network threats and real-world examples

The table below describes five common threats and realistic examples involving familiar apps or services. The examples are illustrative user-level scenarios; they do not claim that the named companies suffered these incidents.

Network threat	What it means	Example involving an app or service
Phishing	A deceptive message or website tricks a person into revealing passwords, OTPs, or payment details.	A user receives a fake Instagram security alert by SMS or email and enters their login details on a look-alike page.
Distributed denial-of-service (DDoS)	A large number of devices send traffic to a service to overload it and make it unavailable.	A food-delivery app such as Zomato could become temporarily inaccessible during a coordinated traffic flood, preventing customers from placing orders.
Man-in-the-middle (MitM)	An attacker secretly intercepts or alters communication between two parties, often on an unsafe network.	A person using public Wi-Fi to access Paytm could be exposed to interception if the connection or device is not properly secured.
Malware	Malicious software can steal information, damage files, or give an attacker control of a device.	A user downloads a fake WhatsApp update from an unofficial website; the installer contains spyware that may access messages or device data.
Password attacks	Attackers try reused, weak, or commonly used passwords through guessing, credential stuffing, or automated attempts.	If an Instagram user reuses a password exposed in another service's breach, an attacker may try the same credentials to access the account.

Question 2. Brute-force simulation for a 4-digit PIN

A brute-force attack tries every possible PIN until it finds a match. The following educational Python program checks all values from 0000 to 9999 against a preset PIN and prints the number of attempts. It runs locally against a fixed demonstration value; it does not connect to or attack any real account or device.

```
# Educational offline brute-force demonstration
preset_pin = "4827"
attempts = 0
found = False

for number in range(10000):
    guess = f"{number:04d}"   # Keep leading zeroes (0000 to 9999)
    attempts += 1
```

```
    if guess == preset_pin:
        found = True
        print("PIN guessed:", guess)
        print("Number of attempts:", attempts)
        break

if not found:
    print("PIN was not found.")
    print("Number of attempts:", attempts)
```

Explanation and sample output

The loop tests each four-digit string in ascending order. The attempts counter increases once for every PIN checked. Because the demonstration PIN is 4827, the program finds it on attempt 4,828 (counting 0000 as attempt 1).

Sample output

```
PIN guessed: 4827\nNumber of attempts: 4828
```

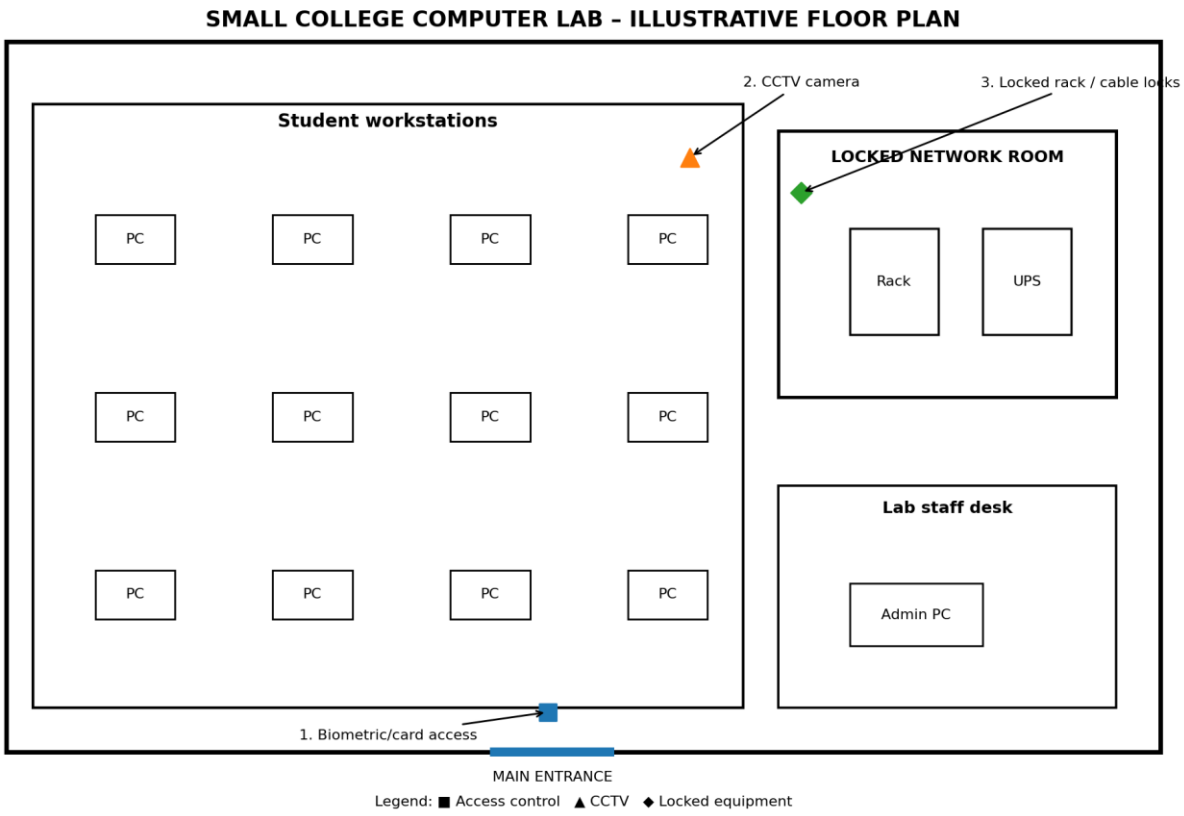
Question 3. Physical security risks in a server room

A server room supporting a popular service such as Zomato or BookMyShow needs protection against physical incidents as well as cyberattacks. Four risks and suitable mitigations are listed below.

Physical risk	Possible impact	One mitigation
Unauthorized entry or theft	A person could steal servers, storage drives, or networking devices, or connect an unauthorized device.	Use locked doors with biometric or access-card entry, maintain access logs, and review them regularly.
Fire or smoke	Fire can destroy equipment and interrupt the service.	Install suitable fire detection and clean-agent fire suppression, with regular inspection and maintenance.
Water leakage or flooding	Water can short-circuit equipment and damage servers and power systems.	Keep equipment off the floor, install leak sensors, and avoid routing water pipes above the server racks where possible.
Power failure or electrical surge	Unexpected shutdowns can interrupt services and damage equipment.	Use appropriately sized UPS units, surge protection, backup power, and periodic tests.

Question 4. Small office / college lab floor plan

The following digital floor plan is an illustrative layout for a small college computer lab. It marks three security measures: controlled entry, CCTV monitoring, and a locked network rack/room.



Security measures shown in the plan

- Biometric or access-card entry: only authorized students, faculty, and technicians should enter the lab; access to the network room should be more restricted.
- CCTV camera: position cameras to cover the entrance and the network equipment area, and retain recordings according to college policy.
- Locked rack and cable locks: keep switches, routers, and servers in a locked rack or room; secure portable equipment and network cables against casual removal.

Additional good practices include keeping an equipment inventory, escorting visitors, maintaining clear emergency exits, and testing door locks and cameras periodically.

Question 5. Seven-point security checklist for a college computer lab

An AI-generated starting checklist was reviewed and edited to fit a college computer lab, where many students use shared computers and lab staff manage equipment and network access.

1. **Keep systems and software updated.** The lab technician should install operating-system, browser, antivirus, and application updates on a planned schedule and remove unsupported software.
2. **Use strong authentication and least privilege.** Give each student an individual login; require strong passwords for staff and admin accounts; enable multi-factor authentication for important systems; do not share administrator credentials.
3. **Protect the network from common attacks.** Use a managed firewall, secure Wi-Fi with current encryption, separate student devices from staff and server networks, and block unnecessary services and ports.
4. **Train users to recognize phishing and malware.** Show students how to check links and sender addresses, avoid unknown USB drives and downloads, report suspicious pop-ups, and never share OTPs or passwords.

5. **Restrict and monitor physical access.** Keep the network rack and router/switch area locked; limit keys or access cards to authorized staff; maintain visitor logs and use CCTV where permitted.
6. **Protect equipment and the room environment.** Use UPS and surge protection, maintain ventilation and suitable temperature, keep liquids away from equipment, and ensure fire detection and appropriate suppression are available.
7. **Back up data and practise incident response.** Back up lab configurations and important academic data to an approved secure location; test restoration; document who to contact and how to isolate an infected PC or report a lost device.

Conclusion

Network security and physical security work together. Technical controls such as firewalls, updates, and strong authentication help reduce cyber risks, while access control, CCTV, equipment locks, power protection, and fire safety protect the physical infrastructure. A college computer lab should combine these measures with clear user awareness and regular maintenance.